



FUNDAMENTALS OF OPEN SOURCE IT AND CLOUD COM...

LINUX FILESYSTEMS AND FILE MANAGEMENT

Storage Management and Configuration

Formatting a Disk and Adding a Filesystem

To format a new disk and add a filesystem, you will need an additional disk.

Select "Expand" to learn more about possible scenarios for setting up your test environment.

You can set up a test environment...

Using a Virtual Machine (VM) Close ^

The VM management application will allow for the creation of another virtual disk and, if necessary, the virtual adapter to connect the virtual disk.

Using a Real Machine Close ^

In most cases, there is no free space on the installed system to add another partition. As such, experimenting with a real machine may lead to unforeseen challenges. The test environment is to use a USB key. The size of the USB key is not a concern; an old one works fine. Please remember the contents of the USB will be lost.

Using a Virtual Machine and a USB Key Close ^

Most VM environments will allow control transfer from the real machine to the VM when the USB is connected. This is also an excellent option.

Next, we will provide an example using an old USB key on a VMware virtual machine.

1. List the existing disk structures as a baseline:

```
student@ubuntu:~$ sudo lsblk -f | grep -v loop
```

NAME	FSTYPE	LABEL	UUID	FSAVAIL	FSUSE%	MOUNTPOINT
sda						
sda1	vfat		7BE6-4C07	511M	0%	/boot/efi
sda2						
sda5	ext4		d5faafb5-02d9-4b8a-97b5-6798fac19adb	2.2G	71%	/
sdb						
sdc						
sdd						
sr0						

2. Insert the USB key, read and acknowledge any messages that pop up. Then, check the disks again.

```
student@ubuntu:~$ sudo lsblk -f | grep -v loop
```

NAME	FSTYPE	LABEL	UUID	FSAVAIL	FSUSE%	MOUNTPOINT
sda						
sda1	vfat		7BE6-4C07	511M	0%	/boot/efi
sda2						
sda5	ext4		d5faafb5-02d9-4b8a-97b5-6798fac19adb	2.2G	71%	/
sdb						
sdc						
sdd						
sde	iso9660	Fedora-WS-Live-25-1-3	2016-11-15-22-03-09-00			
sde1	iso9660	Fedora-WS-Live-25-1-3	2016-11-15-22-03-09-00	0	100%	/media/student/Fedora-WS-Live-25-1-3
sde2	vfat	ANACONDA	6007-F982			
sde3	hfsplus	ANACONDA	1c2eb970-8839-3b51-9555-1d17a206366b			
sr0						

3. This system may have additional information, but the purpose is to find the newly plugged-in USB disk. Notice in step 2 there is an **sde** disk with several partitions; this is typical of an old USB key.

4. This old USB key has its partitions mounted; notice the **MOUNTPOINT** field. The old partitions need

4. This old USB key has its partitions mounted, notice the **MOUNTPOINT** field. The old partitions need to be unmounted; you can use the following command to unmount old partitions:

```
student@ubuntu:~$ sudo umount /media/student/Fedora-WS-Live-25-1-3
```

5. Do not use the file manager to "eject" the USB key, as the root device (**/dev/sde**) will be removed.